

Pass-through

C.Conlon

Fall 2022

Internal Notes

Retailer and Wholesaler FOC given by:

$$\mathbf{p}^r = \underbrace{\mathbf{p}^w + \mathbf{c}^r}_{\mathbf{mc}^r} - (\mathcal{H}_r \odot \Delta_r(\mathbf{p}^r))^{-1} \mathbf{s}(\mathbf{p}^r)$$

$$\mathbf{p}^w = \mathbf{mc}^w + \left(\mathcal{H}_w \odot \left(\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w} \cdot \Delta_r(\mathbf{p}^r) \right) \right)^{-1} \mathbf{s}(\mathbf{p}^r)$$

- ▶ Δ_r is matrix of (retail) demand derivatives $\frac{\partial \mathbf{s}}{\partial \mathbf{p}}$.
- ▶ $\mathcal{H}_r, \mathcal{H}_w$ ownership matrix $(j, k) = 1$ if both products sold by same retailer/wholesaler.
- ▶ $\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w}$ is the **pass-through matrix** (NEW!)

Challenge: We want $\mathbf{p}^r(\mathbf{p}^w)$ and $\mathbf{mc}^w$ but we only have implicit solution for retailer FOC.

How do we get pass-through?

The **pass-through matrix** $\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w}$ can be obtained in one of two ways:

1. Numerically: perturbing the retailer's marginal costs for each possible choice of k and solving

$$\mathbf{p}^r = \mathbf{mc}^r + e_k - (\mathcal{H}_r \odot \Delta_r(\mathbf{p}^r))^{-1} \mathbf{s}(\mathbf{p}^r)$$

(Use Morrow Skerlos (2011) formulation and solve for every (j, k) pair).

2. Analytic: Use the retailer's FOC and apply the implicit function theorem.

$$f(\mathbf{p}^r, \mathbf{mc}^r) \equiv \mathbf{p}^r - \mathbf{mc}^r - (\mathcal{H}_r \odot \Delta(\mathbf{p}^r))^{-1} \mathbf{s}(\mathbf{p}^r) = 0 \quad (\text{retailer FOC})$$

See Jaffe Weyl (AEJM 2013) or Miller Weinberg (2017 Appendix E) or Conlon Rao (2022).

This is what PyBLP does.

The multivariate IFT says that for some system of J nonlinear equations

$$f(\mathbf{p}^r, \mathbf{p}^w) \equiv [F_1(\mathbf{p}^r, \mathbf{p}^w), \dots, F_J(\mathbf{p}^r, \mathbf{p}^w)] = [0, \dots, 0]$$

with J endogenous variables $\mathbf{p}^r$ and J exogenous parameters $\mathbf{p}^w$.

$$\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w} = - \left(\begin{array}{ccc} \frac{\partial F_1}{\partial p_1^r} & \cdots & \frac{\partial F_1}{\partial p_J^r} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_J}{\partial p_1^r} & \cdots & \frac{\partial F_J}{\partial p_J^r} \end{array} \right)^{-1} \cdot \underbrace{\left(\begin{array}{c} \frac{\partial F_1}{\partial p_k^w} \\ \vdots \\ \frac{\partial F_J}{\partial p_k^w} \end{array} \right)}_{=-\mathbb{I}_J} \quad (\text{PTR})$$

Because the system of equations is additive in $\mathbf{mc}^r = \mathbf{c}^r + \mathbf{p}^w$ this simplifies dramatically.

Use the substitution $\Omega(\mathbf{p}^r) \equiv \mathcal{H}_r \odot \Delta_r(\mathbf{p}^r)$, and differentiate the wholesalers' system of FOC's with respect to p_l , to get the $J \times J$ matrix with columns l given by:

$$\frac{\partial f(\mathbf{p}^r, \mathbf{p}^w)}{\partial p_l^r} \equiv e_l - \Omega^{-1}(\mathbf{p}^r) \left[\mathcal{H}_r \odot \frac{\partial \Delta(\mathbf{p}^r)}{\partial p_l^r} \right] \Omega^{-1}(\mathbf{p}^r) \mathbf{s}(\mathbf{p}^r) - \Omega^{-1}(\mathbf{p}^r) \frac{\partial \mathbf{s}(\mathbf{p}^r)}{\partial p_l^r}. \quad (1)$$

The complicated piece is the demand Hessian: a $J \times J \times J$ tensor with elements (j, k, l) ,

$$\frac{\partial^2 s_j}{\partial p_k^r \partial p_l^r} = \frac{\partial^2 \mathbf{s}}{\partial \mathbf{p}^r \partial p_l^r} = \frac{\partial \Delta(\mathbf{p}^r)}{\partial p_l^r}.$$

This also shows a key relationship between **pass through** and **demand curvature** (2nd derivatives).

Pass-through Counterfactuals?

How do we solve for p^w under a counterfactual pass-through matrix?

- Idea: pass-through only augments the matrix $\Delta_r(\mathbf{p}^r)$.
- Example: a constant sales tax rate $P \equiv \frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w} = \text{diag}(1 + \tau_r)$

$$\mathbf{p}^w = \mathbf{m}\mathbf{c}^w + \left(\mathcal{H}_w \odot \left(\frac{\partial \mathbf{p}^r}{\partial \mathbf{p}^w} \cdot \Delta_r(\mathbf{p}^r) \right) \right)^{-1} \mathbf{s}(\mathbf{p}^r)$$

Adapt the Morrow Skerlos ζ fixed point where $P\Delta(\mathbf{p}_t) = P\Lambda_t(\mathbf{p}_t) - P\Gamma_t(\mathbf{p}_t)$

$$\begin{aligned} \mathbf{p}_t &\leftrightarrow \mathbf{c}_t + \boldsymbol{\zeta}_t(\mathbf{p}_t) \quad \text{where} \\ \boldsymbol{\zeta}_t(\mathbf{p}_t) &= \Lambda_t(\mathbf{p}_t)^{-1} \mathbf{P}^{-1} [\mathcal{H}_t^* \odot \mathbf{P} \Gamma_t(\mathbf{p}_t)] (\mathbf{p}_t - \mathbf{c}_t) - \Lambda_t(\mathbf{p}_t)^{-1} \mathbf{P}^{-1} \mathbf{s}_t(\mathbf{p}_t) \end{aligned}$$

For diagonal P (not sure about general case with Hadamard product):

$$\boldsymbol{\zeta}_t(\mathbf{p}_t) = \Lambda_t(\mathbf{p}_t)^{-1} [\mathcal{H}_t^* \odot \Gamma_t(\mathbf{p}_t)] (\mathbf{p}_t - \mathbf{c}_t) - \Lambda_t(\mathbf{p}_t)^{-1} \mathbf{P}^{-1} \mathbf{s}_t(\mathbf{p}_t)$$